

3D printable robots for homes

Researchers from the Massachusetts Institute of Technology (MIT), University of Pennsylvania and Harvard University are developing a desktop technology that would make it possible for the common man to have his own robot. The five-year project, *An Expedition in Computing Printable Programmable Machines*, will help you design, customise and print a specialised robot in a matter of hours.

The main aim of the project is to trace the method of making robots and simplify it so that an average person can understand the design and print a robot as per his specific needs within 24 hours. The team is currently working on making 3D devices from sheets of paper to allow individuals to design and build functional robots from materials as easily accessible as a sheet of paper.

Rehabilitation device for Carpal Tunnel Syndrome patients

Re-Hand, a software-assisted hand rehabilitation device, allows Carpal Tunnel Syndrome (CTS) patients to gain back strength, dexterity and coordination in their fingers and hands. CTS is pressure on the median nerve—the nerve in the wrist that supplies feeling and movement to parts of the hand.

The device, developed by students at Georgia Tech, has three components: a handle to measure total grip strength as well as individualised finger strength; a locking-base platform to allow exercise in the form of wrist flexion and extension; and interactive software that has a game to make the workout fun and a data exportation feature that allows the doctor to monitor the patient's progress. The gaming component makes the boring and difficult process of rehabilitation fun for the patient. The data exportation component allows



Personalised robot made from paper

the doctor to monitor how the patient is faring.

The team has priced the product at \$100. It is now developing a marketing plan and will start by introducing the device to doctors and physical therapists, who will then recommend it to patients.

Oscillating gel could make robots feel

A team of researchers has demonstrated the use of a non-oscillating gel that can be resuscitated in a manner similar to medical cardiopulmonary resuscitation. The team from the University of Pittsburgh and the Massachusetts Institute of Technology (MIT) is researching the behaviour of Belousov-Zhabotinsky (BZ) gel. First developed in the late 1990s, this material pulsates in the absence of any external stimuli. The gel could even 'beat' like a human heart under certain conditions.

MIT researchers proved that chemical oscillations can be triggered by mechanically compressing the BZ gel beyond a critical stress. According to Balazs, the gel could serve as a small-scale pressure sensor for different vehicles or instruments to see whether they had been bumped, providing diagnostics for the impact on surfaces. "Think of it like human skin, which can provide sig-

nals to the brain that something on the body is deformed or hurt," he added.

New plastics 'bleed' and heal like human skin

Researchers from the University of Southern Mississippi have developed a plastic that 'bleeds' red when it is scratched or cut. The plastic heals itself when exposed to light. The new plastic is said to take its inspiration from the human skin to heal scratches and cuts. Scientists believe that it offers the promise of endowing cell phones, laptops, cars and other products with self-repairing surfaces.

Self-healing plastics rely on embedded healing compounds and can self-repair only once. However, the new genre of plastics can heal itself over and over again.

Professor Marek W. Urban, who reported on the research, opines that there are a wide range of potential applications for plastics with warn-and-self-repair capabilities. Scratches in automobile fenders might be repaired by simply exposing the fender to intense light. Critical structural parts in aircrafts might warn of damage by turning red along cracks so that engineers could decide whether to shine the light and heal the damage, or undertake a complete replacement of the component. There